A057 v0.2.0: Master-Mass Reconstruction Blind Falsifier

Omar Iskandarani

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1 Identification

Catalog ID A057; release v0.2.0; SST Falsifier Framework v1.0.6, profile `multilibrary_gpu`. The framework protocol is fail-closed, whereas scientific gates use diagnostic continuation.

2 Question

On one E013-frozen population of current E011 provider anchors, does a topology-stable finite-core vortex energy and its stationarity-qualified variational spectrum provide a target-free Master-Mass kernel?

3 Hypotheses

The null states that independently sourced carriers do not exhibit a reproducible topology-level energy/dynamical kernel beyond numerical and source variation. The alternative requires reproducible topology structure and protected-holdout generalization before any target reveal.

4 Assumptions

The tested sector is incompressible, inviscid, closed-filament dynamics without reconnection or external forcing. E013 fixes the common carrier population from E011 `STATIC_READY` seeds; E010 only reloads the exact source geometry and is not permitted to select replacement carriers. The physical reference object is the three-dimensional kinetic-energy integral. The operational primary kernel is a regularized slender-filament Hamiltonian; an independent Bishop-frame core-tube quadrature is retained as a second reduced finite-core diagnostic.

5 Symbols

$\mathbf{X}_i(s)$ denotes a closed filament component, a the regularization scale, Γ circulation, H_a the line Hamiltonian, q reduced Bishop-frame coordinates, K_H the energy Hessian and J the first-order Jacobian.

6 Equations

Machine-readable science contract

Research question On one E013-frozen common population of current E011 STATIC_READY provider anchors, does the finite-core Hamiltonian and its local variational diagnostics yield reproducible, topology-sensitive target-free kernels suitable for a Master-Mass reconstruction, without reusing any scientific result from legacy falsifier runs?

Objective Re-run the A057 geometry/energy/Lagrangian/Hamiltonian/Hessian/Jacobian program from source bytes for exactly the carriers frozen by E013. Bind every output row to static_seed_id, carrier_id, provider_group and geometry_sha256; preserve provider sensitivity; apply stricter stationarity authority before promoting spectra; freeze all target-free predictions before reveal. Legacy falsifier outputs are forbidden as evidence and scientific FAIL states do not stop later diagnostics.

H_0 On the common current-PKLSA carrier population, no registered energy/dynamical functional is both numerically converged and reproducible across independent E011 provider anchors, or any apparent spectral structure is confined to nonstationary diagnostic states.

H_1 At least one preregistered finite-core topology kernel is numerically converged and reproducible across the E013-frozen independent provider anchors, while stationarity-qualified local spectra supply additional target-free state information without target-specific fitting.

Registered equations

E1: Canonical local mass-equivalent density scale

$$\rho_m = \frac{\rho_f \, \|\mathbf{v}_{\circlearrowleft}\|^2}{2c^2} \tag{1}$$

Dimensional check: (kg mˆ-3)(mˆ2 sˆ-2)/(mˆ2 sˆ-2) = kg mˆ-3.

E2: Regularized finite-core filament Hamiltonian

$$H_a = \frac{\rho\Gamma^2}{8\pi} \sum_{i,j} \oint_{K_i} \oint_{K_j} \frac{d\mathbf{X}_i \cdot d\mathbf{X}_j}{\sqrt{\|\mathbf{X}_i - \mathbf{X}_j\|^2 + a^2}} \tag{2}$$

Dimensional check: rho Gammaˆ2 times a length has units (kg mˆ-3)(mˆ4 sˆ-2)(m)=J.

E3: Dimensionless line-energy kernel after length-scale normalization

$$\widehat{H}_K = \sum_{i,j} \oint \oint \frac{d\widehat{\mathbf{X}}_i \cdot d\widehat{\mathbf{X}}_j}{\sqrt{\|\widehat{\mathbf{X}}_i - \widehat{\mathbf{X}}_j\|^2 + \widehat{a}^2}} \tag{3}$$

Dimensional check: All hatted lengths are dimensionless, so the complete kernel is dimensionless.

E4: Self/mutual coherence matrix

$$C_{ij} = \frac{E_{ij}}{\sqrt{|E_{ii}E_{jj}|}} \tag{4}$$

Dimensional check: Energy divided by energy is dimensionless.

E5: Reduced conservative Lagrangian

$$L(q, \dot{q}) = \tfrac{1}{2}\dot{q}^{\mathsf{T}} M \dot{q} - U(q) \tag{5}$$

Dimensional check: The blind reduced lane is nondimensionalized; after restoration both terms carry energy units.

E6: Reduced Hamiltonian from the Legendre transform

$$H(q,p) = \tfrac{1}{2} p^{\mathsf{T}} M^{-1} p + U(q), \qquad p = M\dot{q} \tag{6}$$

Dimensional check: The blind reduced lane is dimensionless; after restoration H has energy units.

E7: Energy Hessian at a registered carrier geometry

$$K_H = \nabla_q^2 U(q)\big|_{q=0} \tag{7}$$

Dimensional check: Dimensionless in the normalized lane.

E8: First-order dynamical Jacobian

$$J = \begin{pmatrix} 0 & I \\ -M^{-1}K_H & -M^{-1}G \end{pmatrix} \tag{8}$$

Dimensional check: In normalized time J is dimensionless per unit normalized time; a chosen physical time scale restores s^-1.

E9: Spectral ratio with no protected target embedded

$$q_{21}(K) = \omega_2(K)/\omega_1(K) \tag{9}$$

Dimensional check: Ratio of equal-frequency units is dimensionless.

E10: SI mass prediction from the regularized Hamiltonian

$$M_K^{(\rho)} = H_K^{(\rho)}/c^2 \tag{10}$$

Dimensional check: J/(m^2 s^-2)=kg.

E11: Three-dimensional kinetic-energy mass functional used as the physical reference object

$$M_K^{(3D)} = \frac{1}{2c^2} \int_{\mathbb{R}^3} \rho_K(\mathbf{x}) \|\mathbf{v}_K(\mathbf{x})\|^2 \, d^3x \tag{11}$$

Dimensional check: (kg m^-3)(m^2 s^-2)(m^3)/(m^2 s^-2)=kg.

E12: Core-tube quadrature kernel used as an independent reduced check on the line Hamiltonian

$$\widehat{E}_{\text{tube}} = \frac{1}{2} \int_{\widehat{\Omega}_{\text{tube}}} \|\widehat{\mathbf{v}}(\widehat{\mathbf{x}})\|^2 \, d^3\widehat{x}, \qquad E_{\text{tube}} = \frac{\rho \Gamma^2 r_c}{32\pi^2} \widehat{E}_{\text{tube}} \tag{12}$$

Dimensional check: The hatted integral is dimensionless; rho Gamma^2 r_c has units J.

Registered code-to-mathematics steps

1. **S1** [G1; equations E3]: Verify the E013 common carrier manifest, bind to E011 STATIC_READY provider anchors, reload only those exact source bytes through E010 parsers, and recheck raw/geometry hashes.

2. **S2** [G2; equations E1, E2, E3]: Run analytic/synthetic controls, SI scale checks, orientation/rigid-motion/resampling controls.

3. **S3** [G3; equations E3]: Compute length, reach-normalized ropelength proxy, curvature, torsion, writhe/linking, Bishop holonomy, and convergence statistics.

4. **S4** [G4; equations E3]: Evaluate held-out independent carrier groups without pooling distinct carriers as numerical refinement levels.

5. **S5** [G5; equations E2, E3]: Compare C++/pybind11 FP64 regularized energy matrix with Python/NumPy FP64 reference.

6. **S6** [G6; equations E2]: Record framework SYCL/DD32 screening evidence separately from certification authority.

7. **S7** [G7; equations E3]: Quantify replication across E013-frozen provider groups; never replace missing providers with legacy carriers.

8. **S8** [G8; equations E2, E4, E11, E12]: Compute self/mutual energy matrices, coherence spectra, and an independent Bishop-frame core-tube kinetic-energy quadrature for a bounded topology subset.

9. **S9** [G10; equations E5]: Measure normalized energy-gradient stationarity on each common carrier using the profile-frozen v0.2.0 threshold; nonstationary states remain diagnostic only.

10. **S10** [G11; equations E5, E6]: Evaluate Legendre-transform closure and reduced Hamiltonian conservation.

11. **S11** [G12; equations E7]: Construct finite-difference Hessians on the same common carriers and promote stability interpretation only for stationarity-qualified states.

12. **S12** [G13; equations E8, E9]: Construct Jacobians, extract spectra, and compare oscillatory frequencies with direct reduced integration.

13. **S13** [G14; equations E9]: Certify target-free spectral ratios only across repeated stationarity-qualified provider anchors of the same topology.

14. **S14** [G15; equations E3]: Cross-validate simple geometry surrogates against the finite-core energy kernel with topology-grouped holdouts.

15. **S15** [G16; equations E3, E9]: Compare preregistered energy-only and geometry-plus-spectrum surrogate families under a complexity penalty.

16. **S16** [G17; equations E3, E10]: Freeze a complete target-free topology prediction table and anonymous composite predictions before reveal.

17. **S17** [G18; equations E3]: Evaluate protected topology holdout generalization internal to the geometry/energy task; no observed mass targets are used.

18. **S18** [G19; equations E2, E10]: Restore SI scales using canonical SST constants in two density lanes and write target-free absolute predictions.

19. **S19** [G20; equations E2, E7, E8]: Stress core regularization, discretization, carrier selection, and perturbation amplitude; quantify sensitivity.

20. **S20** [G21; equations E3, E9]: Jackknife across provider/lineage groups and report whether findings replicate in independent evidence families.

21. **S21** [G22; equations E3, E5, E6, E8, E10]: Assign a blind multi-level classification from all recorded evidence while preserving diagnostic results from failed upstream scientific gates.

7 Algorithm

Carriers are reloaded from source-native bytes, independently resampled, reach-normalized, and evaluated on preregistered resolution/core ladders. Geometry, self/mutual line energy, an independent core-tube kinetic-energy quadrature, coherence, reduced-action residuals, Hessians and Jacobians are computed without protected targets. Topology-grouped model competition predicts the energy kernel rather than observed masses.

8 Sources

The carrier population is frozen upstream by E013 from E011-v0.3.0 `STATIC_READY` provider anchors. E010-v0.3.1 is used only to reload the exact source-native bytes and optional topology metadata for those already selected carriers. Every row is bound to `static_seed_id`, `carrier_id`, `provider_group`, and `geometry_sha256`. Synthetic controls are implementation evidence only.

9 Gates and diagnostic continuation

Only G0 is a hard framework prerequisite. Every scientific gate depends directly on G0 rather than on the preceding scientific gate. Consequently a scientific FAIL does not prevent later computation. A separate soft-dependency ledger records interpretive caveats. This distinction is intentional: evidence is retained while unsupported promotion is prevented.

10 Numerics

BASIC uses one E013 provider anchor per available topology; FULL and CERTIFY use the same E013-frozen provider-anchor identities, with CERTIFY increasing numerical resolution and perturbation authority rather than silently changing the carrier population. Regularized double line integrals are evaluated in FP64. A separate tube-support velocity quadrature probes the three-dimensional kinetic-energy route on a bounded subset. Hessians use centered finite differences under small Bishop-frame perturbations.

11 Backends

Python/NumPy FP64 is REFERENCE. C++/pybind11 FP64 is CERTIFICATION and must satisfy relative-L2 $\leq 10^{-10}$. SYCL FP32 and DD32/FP32x2 remain SCREENING_ONLY under framework v1.0.6 and cannot certify the scientific result.

12 Blindness

Protected observed targets, protected human identities and protected historical numerical comparison values reside only under `private/`. The blind run freezes a complete topology prediction table first. Reveal requires the v1.0.6 output-integrity and nonced-commitment checks.

13 Statistics

Primary diagnostics are within-topology coefficient of variation, between/within spread ratio, independent-group replication fraction, topology-grouped confirmation/holdout log-RMSE, carrier jackknife sensitivity, core-law sensitivity and target-free spectral-ratio stability.

14 Results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	10edcbc92ca37f8edc885895a5aadfb281e2eedcbb8d338d46646b521234146d
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	de44c65dd6e24c90579ae3b0274f1713e332968405faa22171aa962a02ea9850

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Is the run exactly bound to an E013 common-carrier manifest, and do all E011 STA
G2	PASS	Do analytic controls, SI dimensional scales, rigid-motion/orientation/resampling cont
G3	UNRESOLVED	On the common carriers, is the regularized energy kernel numerically converged, and
G4	UNRESOLVED	Do held-out confirmation topologies with repeated E013 provider anchors reproduce
G5	PASS	Does strict native C++ FP64 reproduce Python/NumPy FP64 for the regularized en
G6	PASS	If available, does framework GPU/DD32 screening evidence satisfy its frozen screeni
G7	UNRESOLVED	Does the energy kernel replicate across the independent E011 provider anchors froze
G8	PASS	For multicomponent carriers, does self+mutual energy reconstruct total energy and y
G10	FAIL	Are the source-native common carriers stationary enough under the reduced energy t
G11	PASS	Does the Legendre transform close and is the reduced Hamiltonian conserved to the
G12	FAIL	Are Hessians symmetric/converged, with stability authority restricted to stationarity
G13	PASS	Does the derived Jacobian spectrum agree with direct reduced integration for resolva
G14	UNRESOLVED	Are target-free spectral ratios reproducible within topology across repeated stationar
G15	PASS	Which preregistered geometry-only surrogate best predicts the finite-core energy kern
G16	FAIL	Does adding coherence/spectral information improve protected energy-kernel general
G17	PASS	Has a complete target-free topology prediction table, including anonymous registered
G18	PASS	Do the selected blind model families generalize to a protected topology holdout not u
G19	PASS	Can absolute SI predictions be generated from the frozen dimensionless kernel in bot
G20	PASS	Are primary conclusions robust to core regularization, discretization, carrier jackknif
G21	UNRESOLVED	Do primary observables survive the exact E013 provider-anchor jackknife and declare
G22	FAIL	What is the blind evidence classification when all diagnostic gates are considered wit
G9	DEFERRED	After a completed blind FULL/CERTIFY run, does commitment-verified reveal pres

Backend manifest


```
{
  "cpp_certification": {
    "authority": "CERTIFICATION",
```

```
    "gate_status": "PASS"
  },
  "diagnostic_continuation": true,
  "gpu_screening": {
    "authority": "SCREENING_ONLY",
    "gate_status": "PASS"
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_gate_dependencies": "soft; see DEPENDENCY_CAVEATS.json"
}
```


Source manifest


```
{
  "manifest_sha256": "7bd1bd0e30b346240832ae834801c181f196dc119e771e8a56eca6f5fafc66a1",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E011_SKLSA_v0.3.0",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s
      "provenance_family": "E011_SKLSA_STATIC_READY",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "static_seed_qualification_authority",
      "sha256": "eda7cdcddaf851d6e83b225df5182634a70b9678bfe3d3a0f338a28c2a0e8668",
      "source_id": "S_E011_STATIC_INDEX",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E011_SKLSA_v0.3.0",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E011_sklsa_s
      "provenance_family": "E011_SKLSA_STATIC_READY",
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        "hits": []
      },
      "role": "provider_anchor_authority",
      "sha256": "999739dad02aa2b4dbc6c00411c9e21cfc59252de5daff65eb6471a168782d7d",
      "source_id": "S_E011_PROVIDER_ANCHORS",
      "version": "v0.3.0"
```

```
    },
    {
      "catalog_id": "E013",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E013_protocol",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E013_PKLSA_C
      "provenance_family": "E013_PKLSA_Cross_Falsifier",
      "registry_hint": {
        "catalog_id": "E013",
        "hits": []
      },
      "role": "cross_falsifier_common_carrier_protocol",
      "sha256": "162a65d09d431ebee4943a1f348770f6f7c442d257815f96d6bcb6cf2377f187",
      "source_id": "S_E013_CAMPAIGN_CONFIG",
      "version": "v0.1.1"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E010_implementation",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p
      "provenance_family": "E010_PKLSA_v0.3.1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "source_native_geometry_parser",
      "sha256": "597d5e3b06b0a91884edd2a7ffa99b2c7160940c68c9da39addb0d0c32796d26",
      "source_id": "S_E010_LOADER",
      "version": "v0.3.1"
    },
    {
      "catalog_id": "A057",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "historical_model_definition",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A057_Master
      "provenance_family": "VAM_to_SST_translation",
      "registry_hint": {
        "catalog_id": "A057",
        "hits": []
      },
      "role": "historical_model_structure_only",
      "sha256": "1acdaf7517d10ccab86de1c12c78212011b6b89e9edaa445fdd4942ed93e31ac",
      "source_id": "S_HISTORICAL_MODEL",
      "version": "v0.2.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```

15 Interpretation

A later diagnostic may remain numerically informative even when an earlier scientific gate fails. Such a result is explicitly downgraded by `DEPENDENCY_CAVEATS.json`; for example a Jacobian around a nonstationary carrier is a local numerical diagnostic rather than a certified stability spectrum. A mass claim requires the separately registered blind prediction and post-reveal comparison.

16 Reproducibility

Run `run_00_install.cmd`, verify/freeze the protocol, then execute BASIC, FULL or CERTIFY. The canonical Workbench root defaults to `C:\workspace\projects\SST-Workbench` and may be overridden by `SST_WORKBENCH_ROOT`.

References

[1] H. K. Moffatt, 1969, "The degree of knottedness of tangled vortex lines," *Journal of Fluid Mechanics* 35, 117–129. DOI: 10.1017/S0022112069000991.

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[3] R. L. Bishop, 1975, "There is More than One Way to Frame a Curve," *American Mathematical Monthly* 82, 246–251. DOI: 10.1080/00029890.1975.11993807.

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